WEC Array PTO Power Analysis

Surrogate Modeling Methodology and Results

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1. Executive Summary

The goal of this work was to produce three outcomes for a two-device wave energy converter (WEC) array. First, the study aimed to estimate PTO power output over a wide range of sea states. Second, it aimed to characterize how that power output varies as the environmental and geometric inputs change. Third, it aimed to produce a computationally efficient surrogate model that could estimate these quantities without repeatedly executing the full hydrodynamic simulation workflow.

To achieve these goals, a WAMIT-based hydrodynamic database was constructed for a two-WEC array over spacing, significant wave height, peak period, and direction. This database was then used to fit a reduced-order polynomial surrogate for PTO total power. The environmental domain used to interpret the surrogate combined two datasets with distinct roles: a large global deep-water dataset derived from NDBC buoys in water depths greater than or equal to 50 m, together with a secondary deep-water wave-condition check, and a North Sea dataset used as a specific climatology embedded within that broader domain.

The final reduced surrogate retained 12 terms from the original third-order polynomial library after ridge regularization and backward elimination. The reduced model achieved validation performance of $\mathrm{RMSE}_{\mathrm{val}} = 4.009$ and $R^2_{\mathrm{val}} = 0.927$, showing that it preserved the main predictive structure of the simulation-derived response surface while remaining computationally efficient.

In addition to predicting PTO power, the surrogate was used as an interpretation tool. Analytic derivatives of the reduced model were used to define a total sensitivity measure over the reported (H_s, T_p) domain, making it possible to identify where the surrogate is most responsive to changes in the governing inputs. These results showed that the largest sensitivities occur in the high-H_s, low-T_p portion of the domain and that the dominant contributions arise from H_s-driven terms, especially the linear, quadratic, and mixed height-period terms. The surrogate therefore provides not only a fast predictor of PTO response, but also a compact and physically interpretable description of how that response varies across sea states.

2. Data Used in This Study

Two environmental datasets were used in this study, each serving a different purpose. The first was a large global deep-water dataset constructed from the NOAA National Data Buoy Center (NDBC). The second was a North Sea dataset, which was used as a specific site

climatology against which the surrogate response surfaces could be interpreted. Together, these datasets allowed the study to distinguish between a broad physics-based environmental domain and a particular regional wave climate.

2.1. Global deep-water dataset from NDBC

The primary large-scale dataset used in this study was constructed from NDBC buoys located in water depths greater than or equal to 50 m. This selection is referred to throughout the study as the *DepthGTE50m* dataset. The purpose of this initial depth filter was to exclude shallow-water and nearshore records whose dynamics are strongly shaped by bottom interaction, shoaling, and other finite-depth effects that are not representative of open-ocean deep-water wave behavior.

However, a depth threshold alone is not sufficient to guarantee that every recorded sea state satisfies the deep-water assumption. For that reason, a secondary physics-based check was applied to the wave conditions themselves. The linear dispersion relation,

$$\omega^2 = gk \tanh(kh), \tag{1}$$

relates angular frequency ω, gravitational acceleration g, wavenumber k, and water depth h. The deep-water condition is recovered when

$$kh \gg 1, \tag{2}$$

so that

$$\tanh(kh) \approx 1 \tag{3}$$

and the dispersion relation reduces to its deep-water form,

$$\omega^2 \approx gk. \tag{4}$$

Equivalently, this corresponds approximately to the requirement that the water depth be large compared with the wavelength scale of the sea state. In the present work, the NDBC dataset was therefore filtered first by buoy depth and then by wave-condition consistency with the deep-water assumption. The result is a broad, physics-based dataset representing global deep-water wave conditions rather than a single regional climatology.

This broad environmental coverage is important for the surrogate framework. Because the NDBC deep-water dataset spans a large portion of physically relevant deep-water wave

phenomena, it provides a wide support over the (H_s, T_p) plane. A surrogate trained and interpreted over such a domain is less tied to a single site and is therefore expected to remain useful across multiple candidate climatologies. In this sense, the NDBC dataset provides a global deep-water backbone for the study.

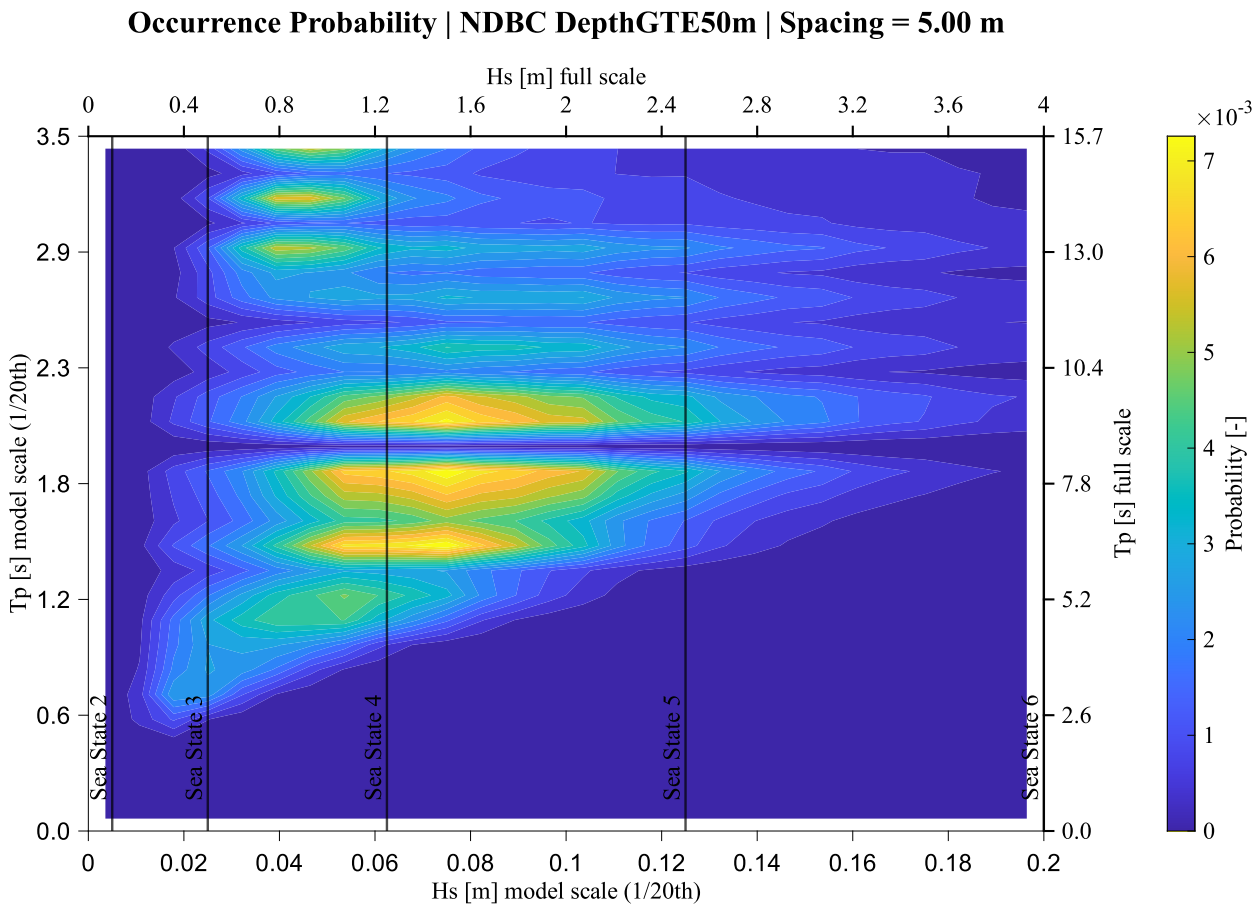

Figure 1: Occurrence probability over the (H_s, T_p) domain for the NDBC DepthGTE50m deep-water dataset at $S = 5.00$ m. This dataset was constructed from NDBC buoys in water depths greater than or equal to 50 m and then filtered to retain sea states satisfying the deep-water-wave assumption.

Figure 1 shows the occurrence probability distribution for the resulting NDBC deep-water dataset. The distribution spans a broad range of significant wave heights and peak periods, illustrating the value of this dataset as a large environmental envelope for surrogate construction and interpretation.

2.2. North Sea climatology

In contrast to the global NDBC dataset, the North Sea dataset was used as a specific climatology. Its purpose was not to define the full deep-water support of the surrogate, but

rather to show how a particular regional wave climate occupies only a subset of that larger environmental domain. This provides a concrete example of how the final surrogate can be applied to an individual climatology after being constructed and interpreted over a broader deep-water space.

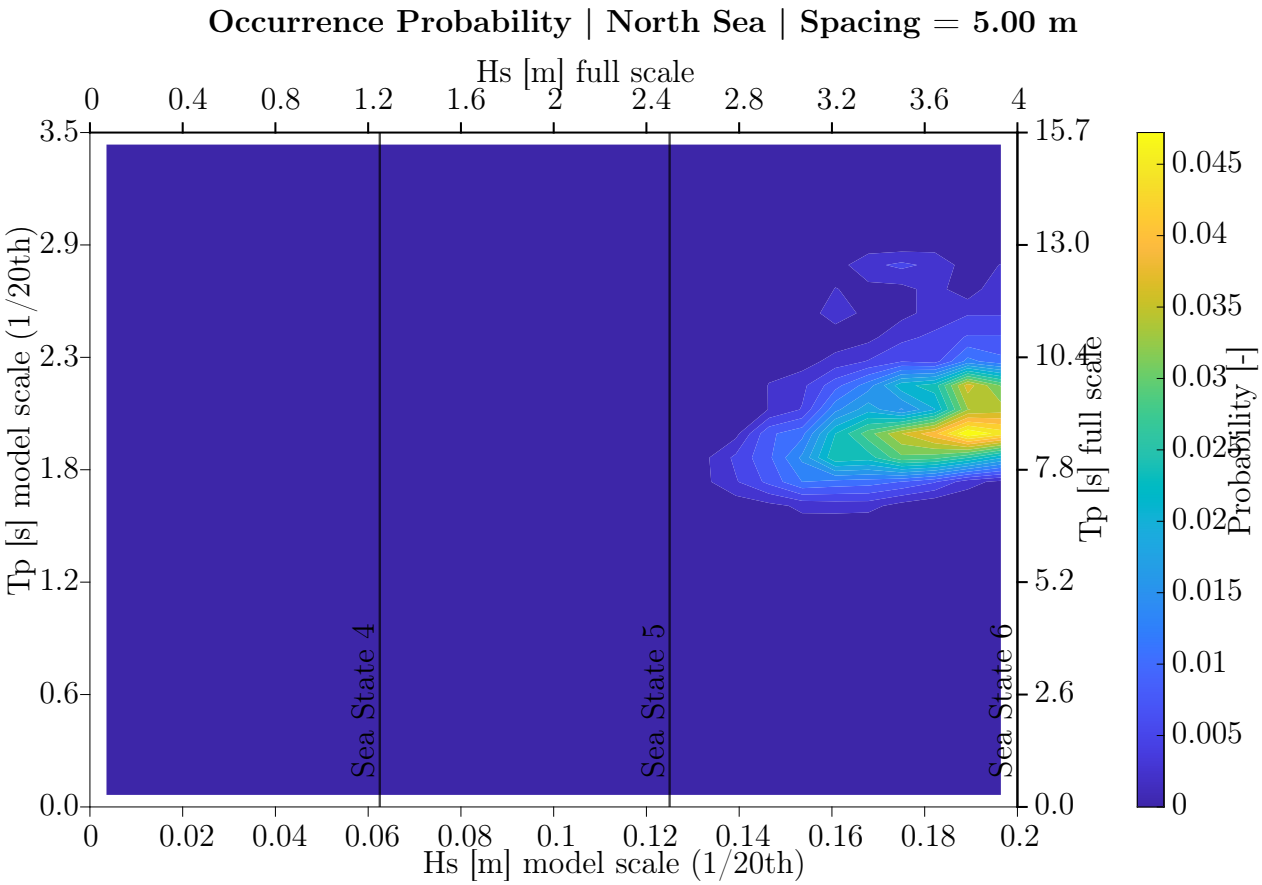

Figure 2: Occurrence probability over the (H_s, T_p) domain for the North Sea dataset at $S = 5.00$ m. Unlike the global NDBC deep-water dataset, the North Sea records occupy a narrower and more localized climatological subset of the full deep-water environmental space.

Figure 2 shows that the North Sea climatology occupies a much narrower region of the (H_s, T_p) plane than the global deep-water NDBC dataset. This distinction is central to the logic of the study. The large NDBC dataset is used to capture most of the deep-water wave phenomena relevant to the surrogate, while the North Sea dataset provides one particular example of a regional climatology embedded within that larger space. The resulting surrogate should therefore be interpreted not as a model for only the North Sea, but as a model intended to remain valid across a wide range of deep-water climatologies, with the North Sea serving as one specific demonstration case.

3. WAMIT Modeling and Simulation Framework

The simulation database used in this work was constructed from a WAMIT-based hydrodynamic model of a two-device wave energy converter array. The float geometry was first built in Blender [1] using device measurements reported by Beringer, Bosma, and Robertson [2]. These measurements were used to reproduce the float dimensions and to define the geometric inputs required for the hydrodynamic analysis.

Once the geometry was constructed in Blender [1], the model was prepared for WAMIT in order to compute the frequency-domain hydrodynamic coefficients required for array analysis. These coefficients provide the linear wave-body interaction quantities needed to describe excitation, radiation, and the resulting array-scale modification of the surrounding wave field over the range of wave conditions considered in this study.

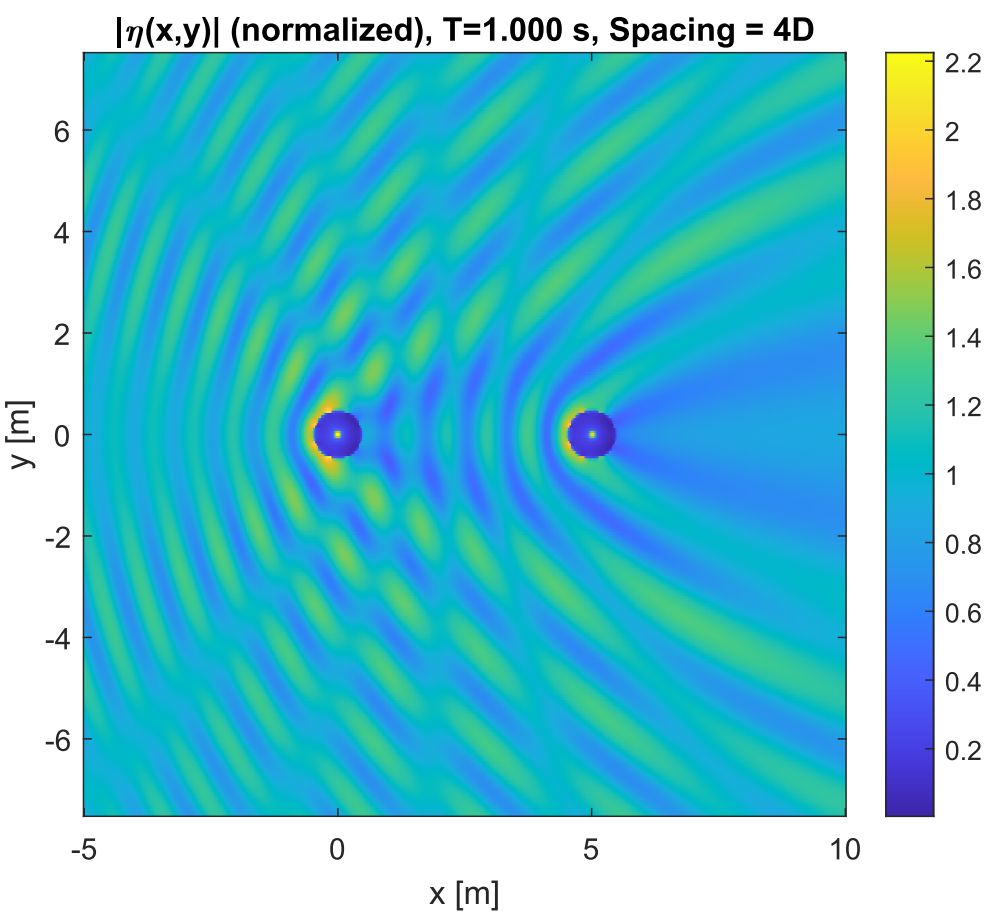

Figure 3: Representative WAMIT wave-field output for the two-WEC array, illustrating the device configuration and the associated shadowing and interaction effects in the surrounding wave field.

Figure 3 provides a representative example of the WAMIT wave-field output used to interpret the array hydrodynamics. In addition to defining the two-WEC configuration, this type of result shows how the presence of one device modifies the local wave environment of the other through scattering and shadowing effects. This is important because the simulation database is intended to capture not only isolated device behavior, but also the hydrodynamic interactions that arise within the array.

The WAMIT results were then used to populate a structured simulation database over the parameter space defined by

$$(S,\ H_s,\ T_p,\ \mathrm{Dir}).$$

By repeating this process over the selected grid of spacings and wave conditions, a lookup table of modeled array response was created. This lookup table forms the foundation of the surrogate-model fitting procedure described in the next section.

The purpose of this workflow is to replace repeated high-cost hydrodynamic simulations with a reduced-order predictive model. Rather than rerunning the full simulation pipeline for each new sea state, the simulation campaign is performed once to generate a representative database, and that database is then used to construct a computationally efficient surrogate.

4. Surrogate Model Construction, Reduction, and Validation

The hydrodynamic response of the wave energy converter (WEC) array was approximated using a polynomial surrogate model. The purpose of the surrogate was to provide a computationally inexpensive approximation of the mapping between environmental conditions and the power absorbed by the power-take-off (PTO) system. This allows rapid evaluation of the array over a large parameter space without repeatedly executing full hydrodynamic simulations.

The surrogate model predicts the mean PTO absorbed power, denoted P_{PTO}, as a function of four environmental and geometric variables:

$$S: \quad \text{device spacing (m)} \tag{5}$$

$$H_s: \quad \text{significant wave height (m)} \tag{6}$$

$$T_p: \quad \text{peak wave period (s)} \tag{7}$$

$$\mathrm{Dir}: \quad \text{mean wave direction (deg).} \tag{8}$$

The resulting input–output map is written as

$$P_{\mathrm{PTO}} = f(S, H_s, T_p, \mathrm{Dir}), \tag{9}$$

where $f(\cdot)$ is approximated by a polynomial expansion fitted to the simulation-derived training data.

Definition of PTO power. The target quantity P_{PTO} is defined as the mean power dissipated through the PTO damping system. For a linear PTO model, the instantaneous absorbed power is given by

$$P_{\mathrm{PTO}} = \frac{1}{2} B_{\mathrm{PTO}} \, |\dot{x}|^2, \tag{10}$$

where B_{PTO} is the PTO damping coefficient and $\dot{x}$ is the velocity response of the device.

The response $\dot{x}$ is obtained from the frequency-domain equation of motion

$$\left[C - \omega^2 (M + A) + i\omega (B_{\mathrm{rad}} + B_{\mathrm{PTO}})\right] x = F_{\mathrm{exc}}, \tag{11}$$

which explicitly includes PTO damping in the response solve. This ensures that the computed P_{PTO} values are physically consistent with the device dynamics and energy dissipation mechanism.

The surrogate model is therefore trained directly on the mean PTO power values computed from the Stage 1 hydrodynamic workflow.

4.1. Input standardization

Before constructing polynomial features, the input variables were standardized in order to improve the conditioning of the regression problem and reduce the numerical imbalance between lower- and higher-order terms. Each raw variable x was transformed to a standardized variable z according to

$$z = \frac{x - \mu}{\sigma}, \tag{12}$$

where μ and σ are the mean and standard deviation computed from the training data.

For the directional variable, a trigonometric representation was used in order to preserve periodicity. The input vector used to form polynomial features was therefore taken as

$$\mathbf{x} = \begin{bmatrix} S & H_s & T_p & \cos(\mathrm{Dir}) & \sin(\mathrm{Dir}) \end{bmatrix}, \tag{13}$$

followed by standardization of each component.

4.2. Polynomial feature expansion

A third-order polynomial expansion was constructed from the standardized input vector $\mathbf{z}$. The surrogate model is expressed as

$$P_{\mathrm{PTO}} = \sum_{i=1}^{N_{\mathrm{terms}}} \beta_i \, \Phi_i(\mathbf{z}), \tag{14}$$

where $\Phi_i(\mathbf{z})$ are polynomial basis functions and β_i are the corresponding regression coefficients.

The feature library includes all monomials up to total degree three, including interaction terms between variables. This polynomial basis is sufficiently rich to capture low-order nonlinearities and interactions between spacing, wave height, period, and direction.

4.3. Ridge-regularized regression

The regression problem was solved using ridge regression, with the objective

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \left(\|\mathbf{y} - \boldsymbol{\Phi}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 \right), \tag{15}$$

where $\mathbf{y}$ is the vector of training PTO power values and $\boldsymbol{\Phi}$ is the polynomial feature matrix.

The role of the regularization term is to control coefficient magnitude, reduce variance, and stabilize the regression in the presence of multicollinearity.

To select an appropriate value of λ, the model was evaluated over a logarithmically spaced grid, and the validation RMSE was computed for each value. The selected λ corresponds to the minimum validation error.

4.4. Backward elimination and model reduction

Although the full polynomial basis captures the PTO power response well, many terms contribute little to predictive accuracy. A backward elimination procedure was therefore applied to obtain a reduced model.

The reduction procedure retained only those polynomial terms that materially influenced predictive performance, as determined by validation RMSE.

4.5. Model validation and acceptance of the reduced model

The predictive performance of the reduced surrogate was evaluated against the full model using an independent validation dataset.

The validation focused on:

- Root-mean-square error (RMSE)
- Parity comparison between predicted and simulated PTO power

The close agreement between the reduced and full models demonstrates that the reduced surrogate preserves the predictive accuracy of the full polynomial expansion while significantly reducing model complexity.

The model comparison in Figure 4 provides a direct visual assessment of the predictive consistency between the full polynomial model and the reduced surrogate model. Each point represents a validation data sample, with the horizontal axis corresponding to the true simulated power output and the vertical axis representing the model prediction. If the surrogate accurately reproduces the simulation results, the points should lie close to the one-to-one reference line. The nearly identical distributions observed for the full and reduced models confirm that eliminating lower-importance polynomial terms does not degrade predictive performance. This comparison therefore validates the reduced surrogate as an accurate and computationally efficient replacement for the full polynomial model.

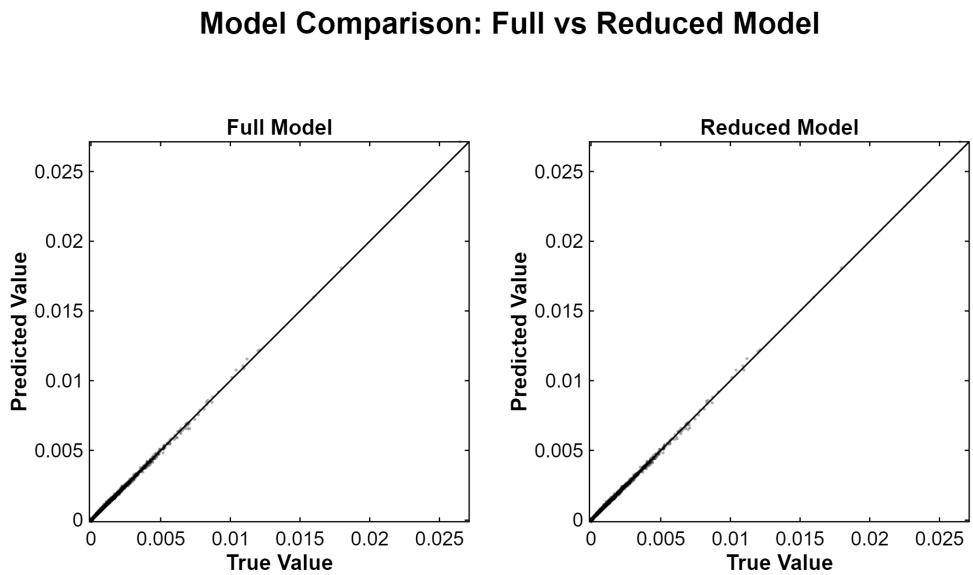

Figure 4: Comparison between the full polynomial model and the reduced surrogate model on the validation set. Each point represents a validation sample, with the horizontal axis corresponding to the simulated PTO power and the vertical axis representing the model prediction. The close agreement of both models with the 1:1 line demonstrates that the reduced model preserves the predictive accuracy of the full model while using fewer polynomial terms.

The model comparison in Figure 4 provides a direct visual assessment of the predictive consistency between the full polynomial model and the reduced surrogate model. The nearly identical distributions observed for the full and reduced models confirm that eliminating lower-importance polynomial terms does not degrade predictive performance. This comparison therefore validates the reduced surrogate as an accurate and computationally efficient representation of the PTO power response surface.

4.6. Reported form of the final reduced model

For reproducibility, the final reduced surrogate may be reported in two equivalent forms. The first is the *standardized-input form*, which is the form used internally during fitting and evaluation. In this representation, each input variable is centered and scaled according to

$$z = \frac{x - \mu}{\sigma}, \tag{16}$$

and the surrogate is written as a polynomial in the standardized variables. This form is numerically convenient and corresponds exactly to the fitted regression problem.

The second is the *dimensional input form*, obtained by expanding the standardized polynomial back into the original variables S, H_s, T_p, cos(Dir), and sin(Dir). This expansion produces an equivalent reduced model written directly in terms of the physical input quantities, and is therefore more convenient for interpretation and reporting. Because the directional input was encoded using cos(Dir) and sin(Dir) to preserve angular periodicity, the reported dimensional model is naturally expressed in those variables rather than as a polynomial directly in the angle measured in degrees.

In the present work, the standardized form is treated as the canonical implementation of the surrogate model, while the dimensional form is reported to provide a more physically interpretable expression for the final reduced approximation.

4.7. Final reduced surrogate used in this study

The final reduced surrogate used in this study was obtained by applying ridge regularization and backward elimination to the full 56-term third-order polynomial library. The retained model contains 12 terms, including the intercept, and the selected ridge parameter was

$$\lambda = 10. \tag{17}$$

The response variable was PTO total power, P_{PTO}, and the predictor set was

$$\left[H_{s\mathrm{model}},\ T_{p\mathrm{model}},\ S,\ \sin\beta,\ \cos\beta\right], \tag{18}$$

where β denotes the equivalent wave heading used in the assigned-power dataset. Direction was represented through $\sin\beta$ and $\cos\beta$ rather than by fitting directly on the angular coordinate, thereby preserving periodicity in the directional input.

The final model was fit on a stratified sampled table containing 444,780 rows drawn from the North Sea, PWS, and Deep Water NDBC datasets. On the validation set, the final reduced surrogate achieved

$$\mathrm{RMSE}_{\mathrm{val}} = 4.00934737357, \qquad R^2_{\mathrm{val}} = 0.927317323246. \tag{19}$$

For reproducibility, the surrogate is reported here in its standardized-input form, which is

the exact form used in fitting and evaluation. Let the raw input vector be

$$\mathbf{x} = \begin{bmatrix} H_{s\mathrm{model}} & T_{p\mathrm{model}} & S & \sin\beta & \cos\beta \end{bmatrix}^{\mathsf{T}}, \tag{20}$$

and let the corresponding standardized variables be

$$\mathbf{z} = \begin{bmatrix} z_{H_s} & z_{T_p} & z_S & z_{\sin\beta} & z_{\cos\beta} \end{bmatrix}^{\mathsf{T}}, \qquad z_j = \frac{x_j - \mu_j}{\sigma_j}. \tag{21}$$

The predictor standardization used in the final model was

$$\mu_{H_s} = 0.107441023793, \qquad \sigma_{H_s} = 0.0615552573873, \tag{22}$$

$$\mu_{T_p} = 2.39326847055, \qquad \sigma_{T_p} = 0.814558171854, \tag{23}$$

$$\mu_S = 5, \qquad \sigma_S = 1, \tag{24}$$

$$\mu_{\sin\beta} = 0.646197864444, \qquad \sigma_{\sin\beta} = 0.309729048094, \tag{25}$$

$$\mu_{\cos\beta} = 0.625108507155, \qquad \sigma_{\cos\beta} = 0.309412381727. \tag{26}$$

The reduced surrogate is written as

$$P_{\mathrm{PTO}} = \phi_{\mathrm{std}}(\mathbf{z})\,\boldsymbol{\beta}_{\mathrm{std}}, \tag{27}$$

where

$$\begin{aligned} \phi_{\mathrm{std}}(\mathbf{z}) = \big[&1,\ z_{T_p},\ z_{H_s},\ z_{T_p}z_{\cos\beta},\ z_{T_p}z_{\sin\beta},\ z_{T_p}^2, \\ &z_{H_s}z_{T_p},\ z_{H_s}^2,\ z_{T_p}z_{\sin\beta}z_{\cos\beta},\ z_{T_p}^3,\ z_{H_s}z_{T_p}^2,\ z_{H_s}^2 z_{T_p}\big], \end{aligned} \tag{28}$$

and

$$\boldsymbol{\beta}_{\mathrm{std}} = \begin{bmatrix} 11.1698527357 \\ -4.97356766869 \\ 12.6679854083 \\ 2.06613448754 \\ 2.03598656242 \\ -2.06074277311 \\ -3.44890470957 \\ 3.80091512988 \\ -1.10694585524 \\ 0.857029022156 \\ -1.36732881429 \\ -1.79544914462 \end{bmatrix}. \tag{29}$$

This standardized form is the authoritative computational representation of the final reduced surrogate used in this study.

Total sensitivity magnitude. Sensitivity analysis in this work is based on analytic derivatives of the surrogate. To compare derivative contributions across variables with different physical units, each partial derivative is scaled by the span of the corresponding variable over the plotted domain. The total sensitivity magnitude is then defined as

$$\mathcal{S}_{\text{tot}} = \sqrt{\left(\left|\frac{\partial P_{\text{PTO}}}{\partial H_s}\right| \Delta H_s\right)^2 + \left(\left|\frac{\partial P_{\text{PTO}}}{\partial T_p}\right| \Delta T_p\right)^2 + \left(\left|\frac{\partial P_{\text{PTO}}}{\partial S}\right| \Delta S\right)^2 + \left(\left|\frac{\partial P_{\text{PTO}}}{\partial \beta}\right| \Delta \beta\right)^2}, \tag{30}$$

where ΔH_s, ΔT_p, ΔS, and $\Delta \beta$ are the variable spans used for the plotted slice. This scaling places the derivative contributions on a comparable footing before combining them into a single total sensitivity measure.

In addition to variable-level sensitivity maps, term-level sensitivity was assessed by applying the same total-sensitivity definition to the derivative contribution of each retained polynomial term separately. This made it possible to compare the retained coefficient magnitudes with the actual derivative-based influence of each term over the reported (H_s, T_p) domain.

5. Results: Surrogate-Predicted PTO Power, Capture Width, and Contribution Fraction

This section presents surrogate-predicted response surfaces at a fixed device spacing of $S = 5.00$ m. As described in the methodology, the hydrodynamic simulation database was used to construct a reduced polynomial surrogate for PTO-related response quantities as functions of spacing, wave height, wave period, and direction. The figures in this section therefore represent evaluations of the fitted surrogate over the (H_s, T_p) plane for the wave-climate domains associated with the NDBC and North Sea datasets, rather than direct pointwise simulation output. The purpose of these figures is to visualize the structure of the learned response surface and to compare how different environmental datasets populate that surface.

In all cases, the horizontal axis is significant wave height in model scale, with a full-scale secondary axis at the top, and the vertical axis is peak period in model scale, with a full-scale secondary axis on the right.

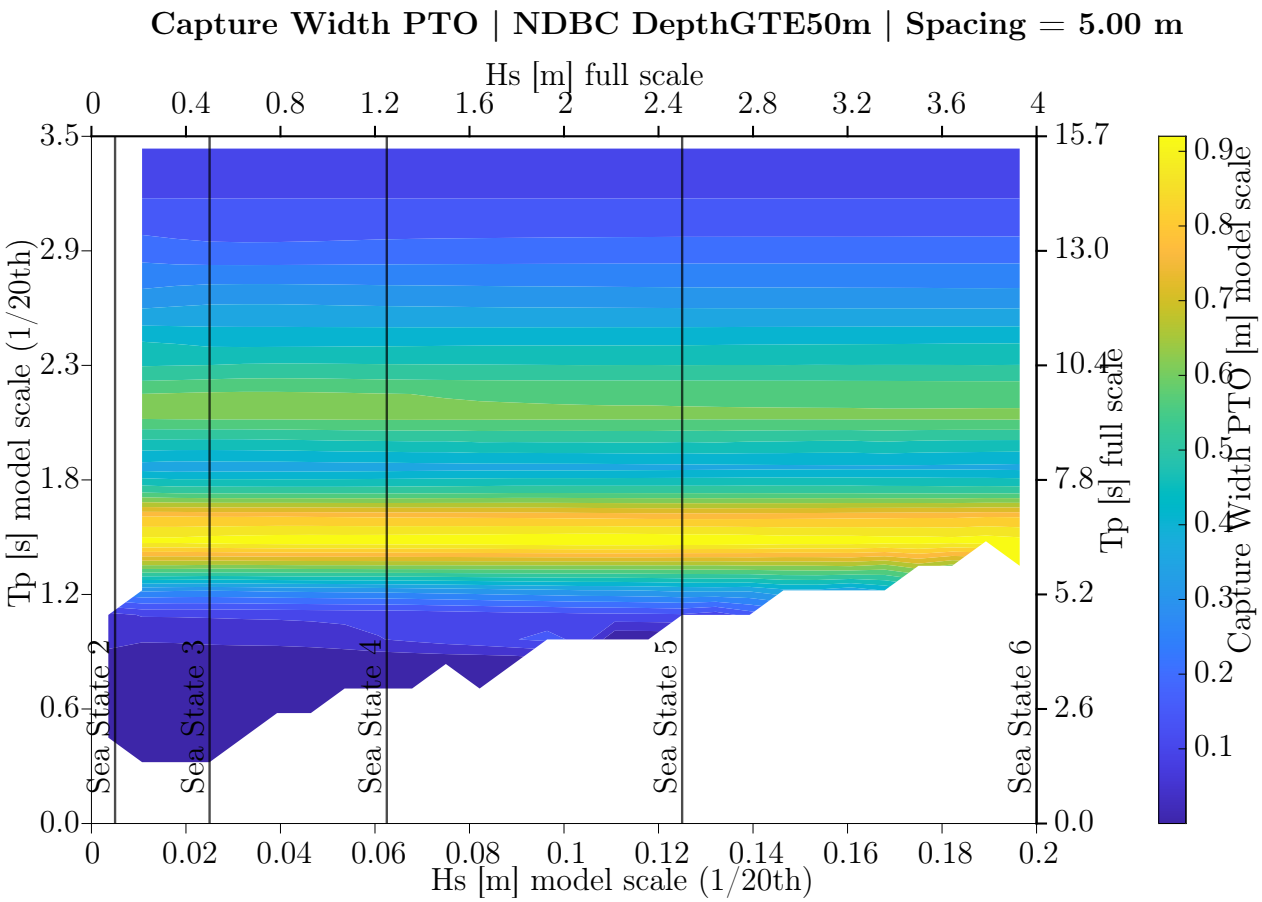

Figure 5: Surrogate-predicted PTO capture width over the NDBC DepthGTE50m wave-climate domain at spacing $S = 5.00$ m.

5.1. NDBC DepthGTE50m

Figure 5 shows the surrogate-predicted capture-width surface over the NDBC DepthGTE50m domain. Because the NDBC dataset spans a broad range of sea states, the surrogate is visualized over a correspondingly broad region of the (H_s, T_p) plane. The dominant feature is a largely horizontal band of elevated capture width centered approximately around $T_p \approx 1.4$–1.6 s in model scale. This indicates that the surrogate identifies a preferred period band for normalized energy extraction at this spacing. Within the admissible region, the dependence on H_s is present but weaker than the period dependence, suggesting that once the incident waves are sufficiently energetic, the dominant efficiency control is period tuning.

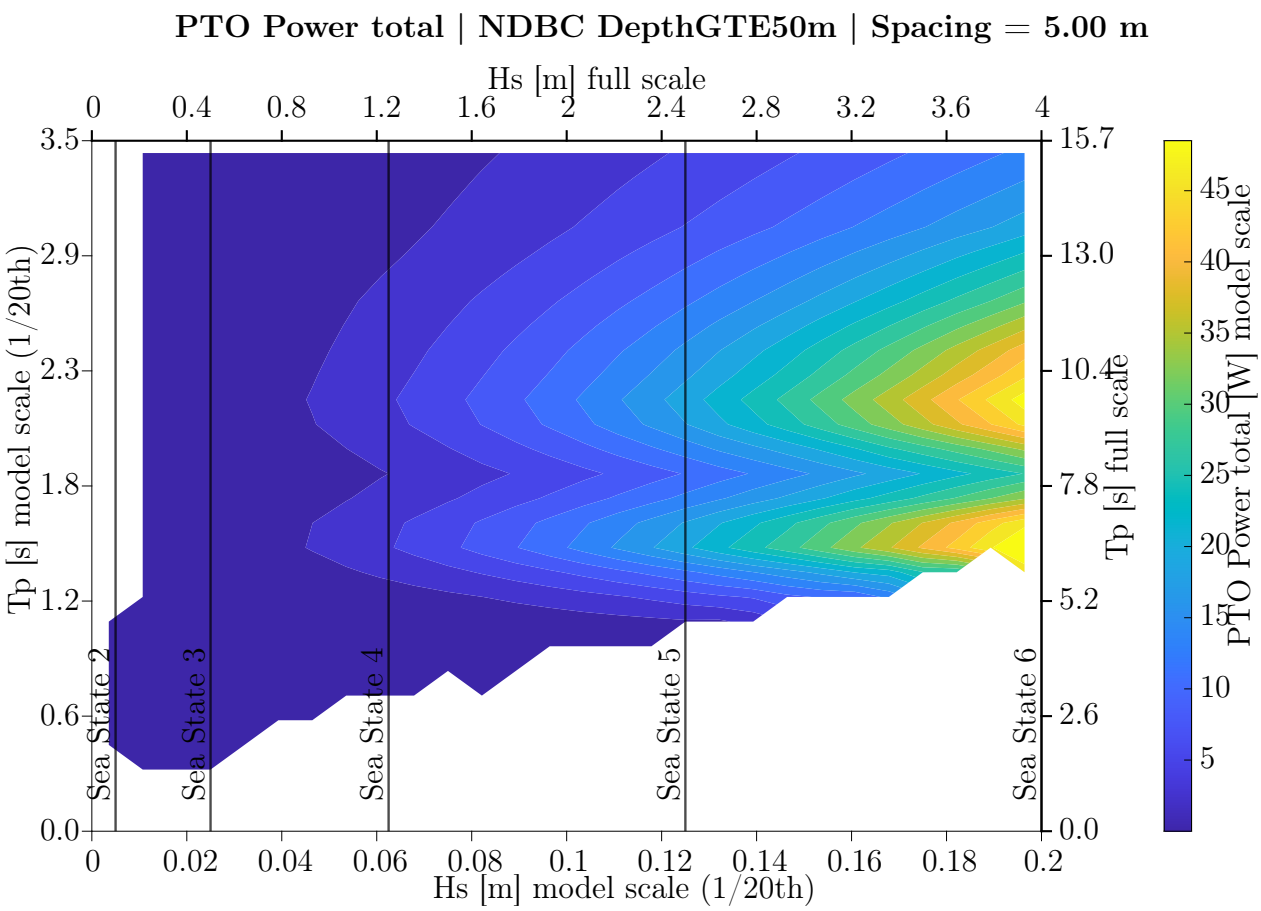

Figure 6: Surrogate-predicted PTO total power over the NDBC DepthGTE50m wave-climate domain at spacing $S = 5.00$ m.

Figure 6 shows the corresponding surrogate-predicted PTO total power surface. In contrast to capture width, the total-power surface increases much more strongly with H_s, reflecting the increase in available incident wave energy as wave height rises. The largest predicted power occurs near the upper-right portion of the occupied domain, where both H_s and T_p are relatively large. The surface also exhibits clear period structure, including a strong response

band around $T_p \approx 1.5$–1.7 s and a secondary high-response region extending toward larger periods as H_s increases. Together, Figures 5 and 6 show that the conditions of highest normalized performance are not identical to those of highest absolute PTO power.

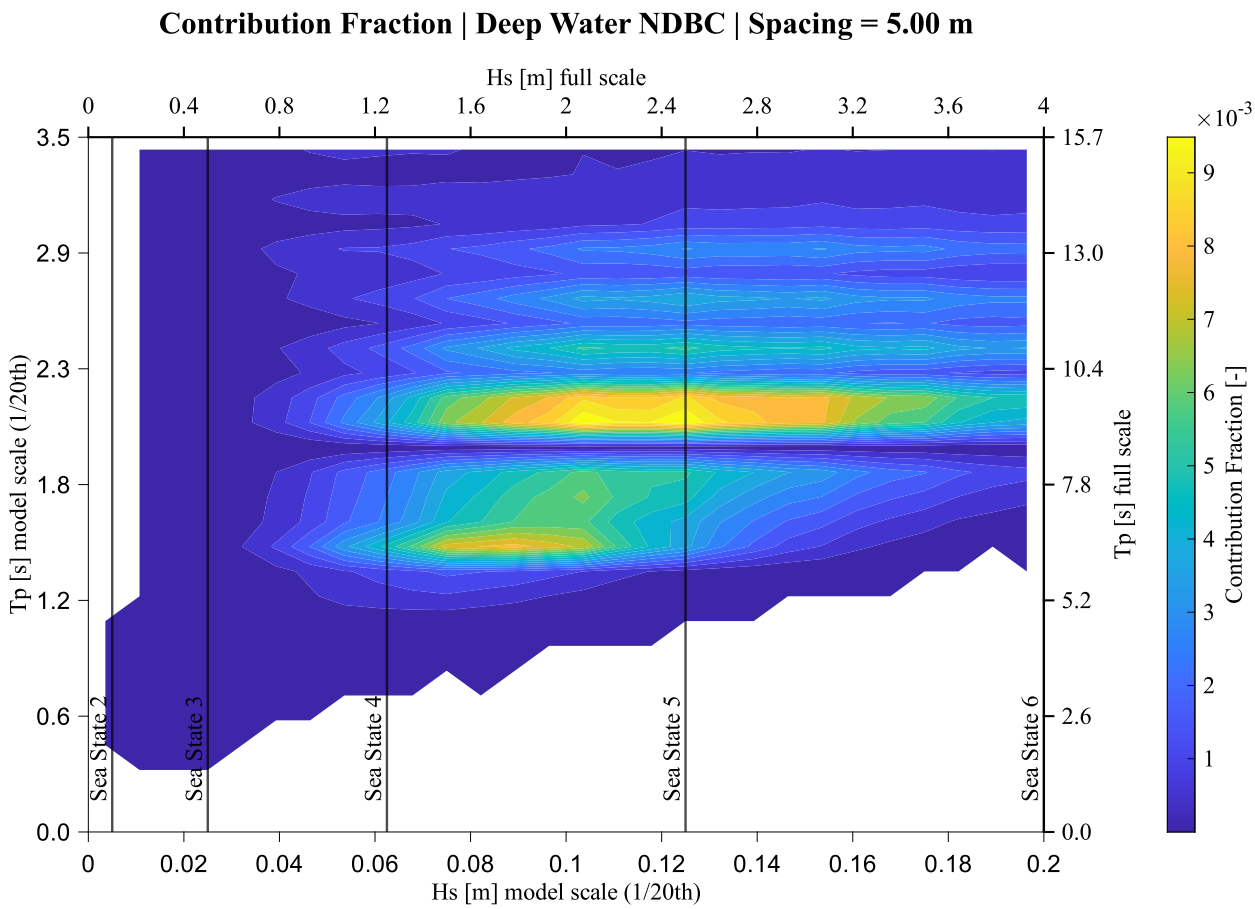

Figure 7: Surrogate-predicted contribution fraction over the NDBC DepthGTE50m wave-climate domain at spacing $S = 5.00$ m.

Figure 7 adds an additional layer of interpretation by showing the surrogate-predicted contribution fraction. This quantity indicates how the total PTO response is partitioned across the array response mechanism represented by the surrogate and is therefore useful for understanding not only how much power is produced, but also how that power is distributed in relative terms across the modeled system contribution. In the NDBC domain, the contribution-fraction surface exhibits a structured dependence on both H_s and T_p, rather than simply mirroring the total-power contours. This indicates that the sea states producing the largest absolute PTO power are not necessarily the same sea states for which the relative contribution pattern is most strongly shifted.

When read together, Figures 6 and 7 suggest that the broad NDBC wave-climate domain contains distinct subregions: some that maximize absolute power, and others that change

the relative contribution balance more strongly. This reinforces the idea that the surrogate is not only recovering a magnitude surface, but also exposing structure in how the system response is partitioned across the environmental domain.

5.2. North Sea

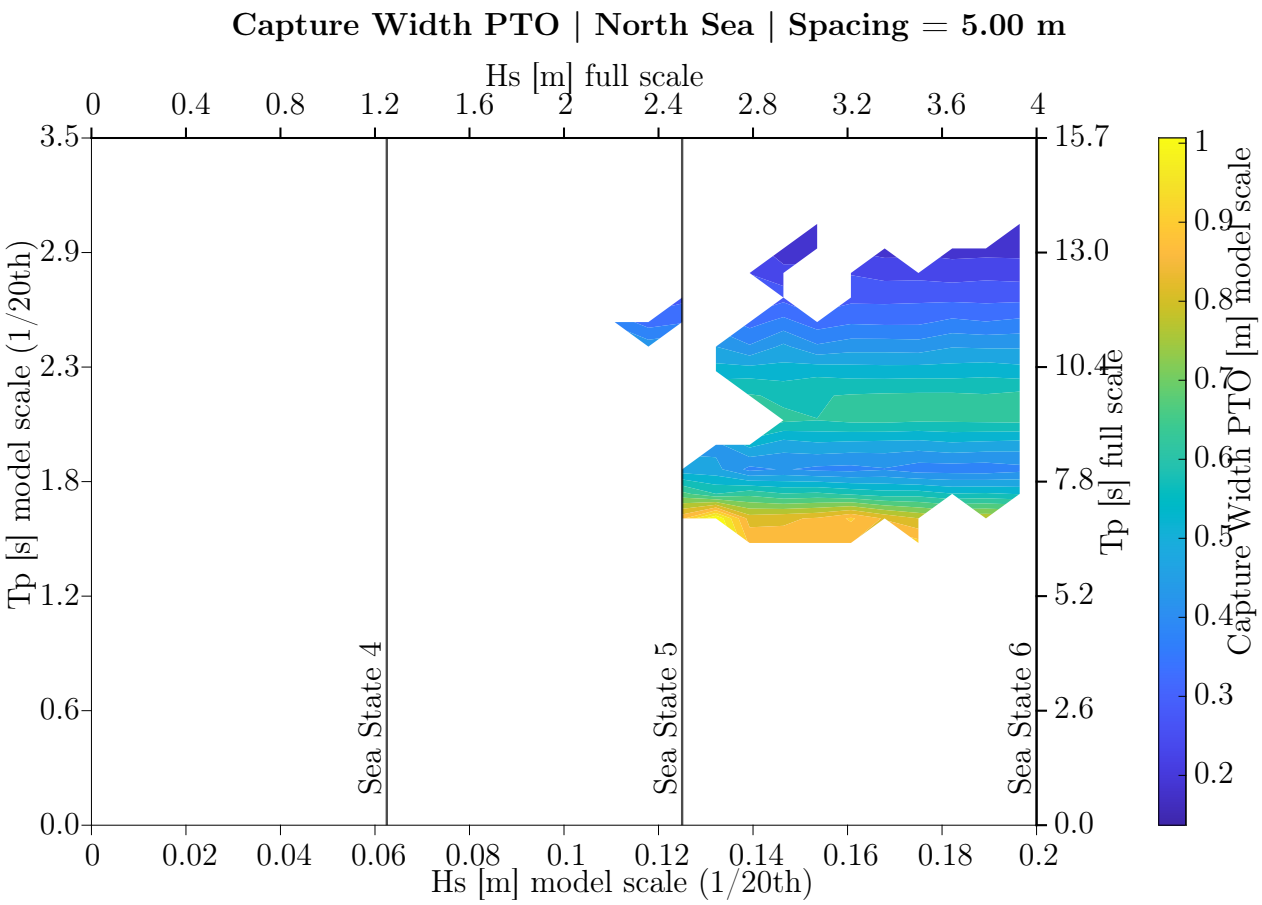

Figure 8: Surrogate-predicted PTO capture width over the North Sea wave-climate domain at spacing $S = 5.00$ m.

Figure 8 shows the surrogate-predicted capture-width surface over the North Sea wave-climate domain. Relative to the NDBC case, the support of the figure is substantially narrower in the (H_s, T_p) plane. This reflects the narrower sea-state region associated with the North Sea dataset. Within this smaller domain, the highest predicted capture-width values again appear near the lower portion of the occupied period band, approximately around $T_p \approx 1.6$–1.8 s in model scale. Capture width then decreases as period increases, indicating that the surrogate continues to identify a preferred efficiency band even under the more restricted North Sea climate.

Figure 9 shows the corresponding surrogate-predicted PTO total power over the North Sea

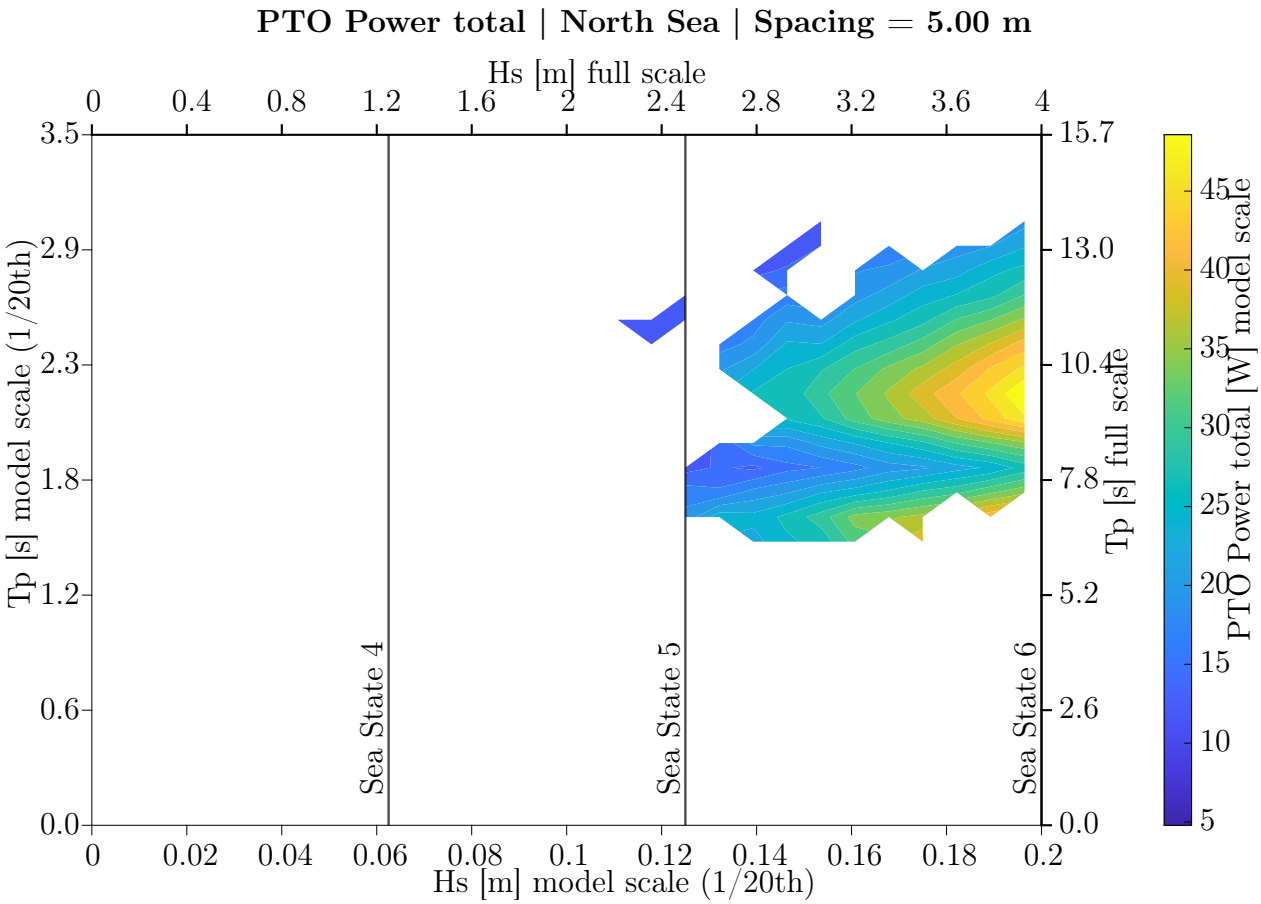

Figure 9: Surrogate-predicted PTO total power over the North Sea wave-climate domain at spacing $S = 5.00$ m.

domain. As in the NDBC case, the largest predicted PTO power occurs near the upper-right edge of the occupied region, where both H_s and T_p are relatively large. The period dependence remains structured and non-monotone, but the response is concentrated over a smaller region than in the NDBC figure because the North Sea climate occupies only a subset of the larger design space.

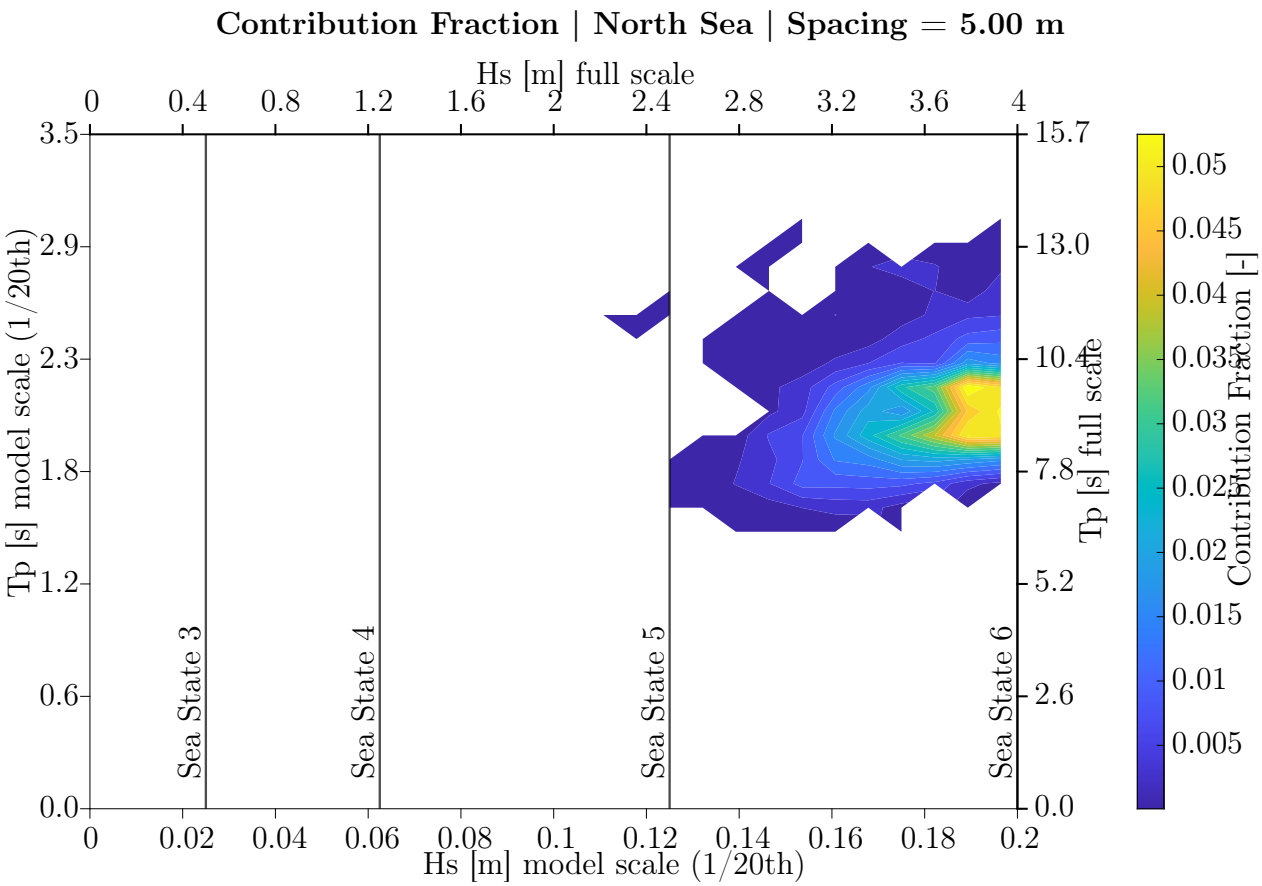

Figure 10: Surrogate-predicted contribution fraction over the North Sea wave-climate domain at spacing $S = 5.00$ m.

Figure 10 shows the surrogate-predicted contribution fraction over the North Sea domain. Because the North Sea support is more localized, the contribution-fraction structure is likewise confined to a smaller region of the response surface. This figure is useful because it shows whether the same region that produces high PTO power also corresponds to a stable or shifted relative contribution pattern. In the North Sea case, the contribution-fraction map remains structured across the occupied domain, indicating that relative contribution is still sensitive to sea-state location even within a narrower climate envelope.

This complements the total-power results. While Figure 9 identifies where the surrogate predicts the largest absolute PTO response, Figure 10 identifies where the relative system

contribution changes most strongly. The two do not need to coincide, and the surrogate makes that distinction visible.

5.3. Comparison Between Datasets

A comparison of the NDBC and North Sea figures reveals three main effects. First, the surrogate surfaces are sampled over very different wave-climate domains. The NDBC dataset spans a broad region of the (H_s, T_p) plane, while the North Sea dataset occupies a more localized subset. Second, in both datasets the capture-width figures emphasize a preferred period band near the lower portion of the occupied T_p range, whereas the PTO total-power figures shift toward larger H_s and somewhat larger T_p. Third, the contribution-fraction figures show that the relative partitioning of the surrogate response is itself structured across the environmental domain and is not reducible to the same patterns seen in total power alone.

This comparison clarifies the analytical value of the surrogate. The surrogate is not merely a compact replacement for the simulation database; it is also an interpretation tool that exposes multiple layers of structure in the learned response surface. For the present spacing of $S = 5.00$ m, the surrogate indicates that the system has a robust period band associated with efficient energy capture, that the largest absolute PTO power is obtained when that favorable structure coincides with sufficiently energetic sea states, and that the relative contribution pattern varies systematically across the same domain. The broader NDBC figures are useful for understanding the full learned response geometry, while the North Sea figures show how a site-specific climate samples only a restricted portion of that geometry.

5.4. Power Sensitivity of the final PTO power surrogate

To interpret how the final reduced surrogate responds to perturbations in its inputs, Figure 11 shows the normalized total sensitivity magnitude, while Figures 12 and 13 show the normalized sensitivity fractions associated with H_s and T_p, respectively. As defined in Section 4.7, the total sensitivity combines the analytic derivatives with respect to H_s, T_p, S, and β after each derivative has been scaled by the span of the corresponding variable over the plotted domain. The resulting field is then normalized by its maximum value over the reported (H_s, T_p) slice, so the figure reports relative sensitivity on a 0 to 1 scale.

Several features are evident. First, the total sensitivity field in Figure 11 is smooth across the domain, which is consistent with the low-order polynomial structure of the final surrogate and indicates that the reported pattern is a property of the fitted response surface itself. Second, the highest normalized sensitivities are concentrated in the lower-right portion of the domain, rather than being distributed uniformly. The strongest response occurs near $H_s \approx 0.33$ to

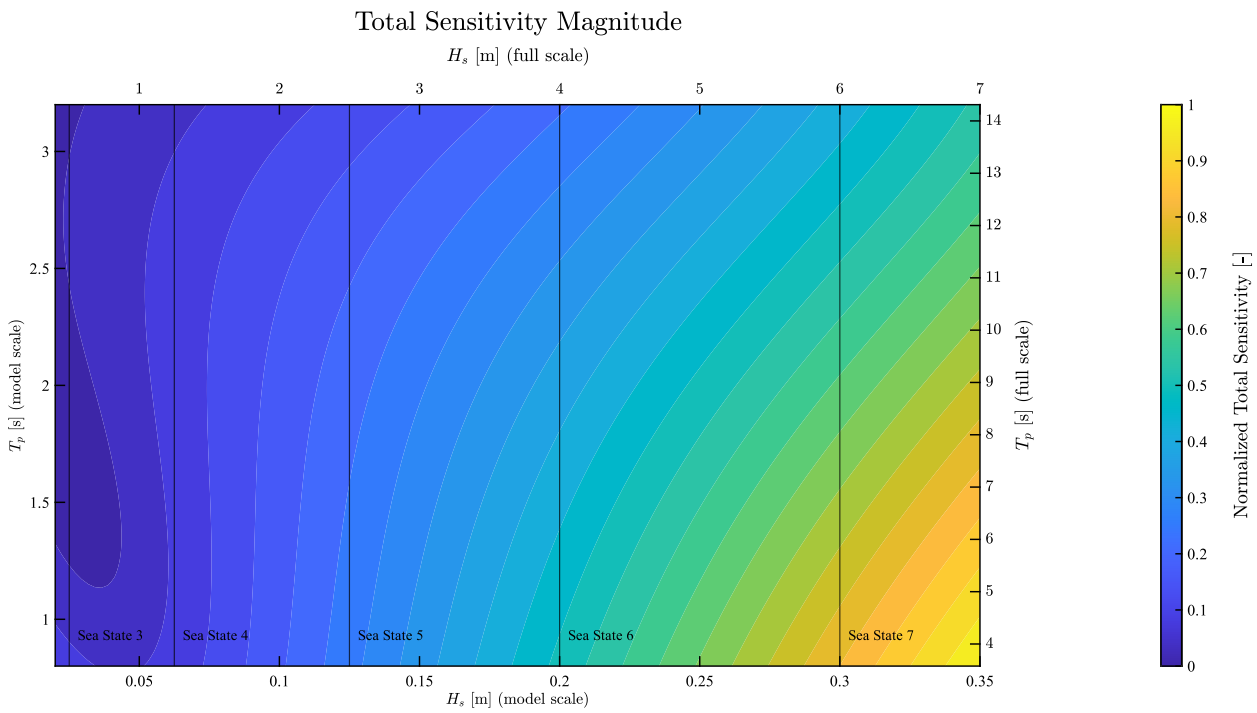

Figure 11: Normalized total sensitivity magnitude of the final reduced PTO power surrogate over the reported (H_s, T_p) domain at fixed spacing and direction. Sea-state boundaries are shown in model scale using 1:20 Froude scaling.

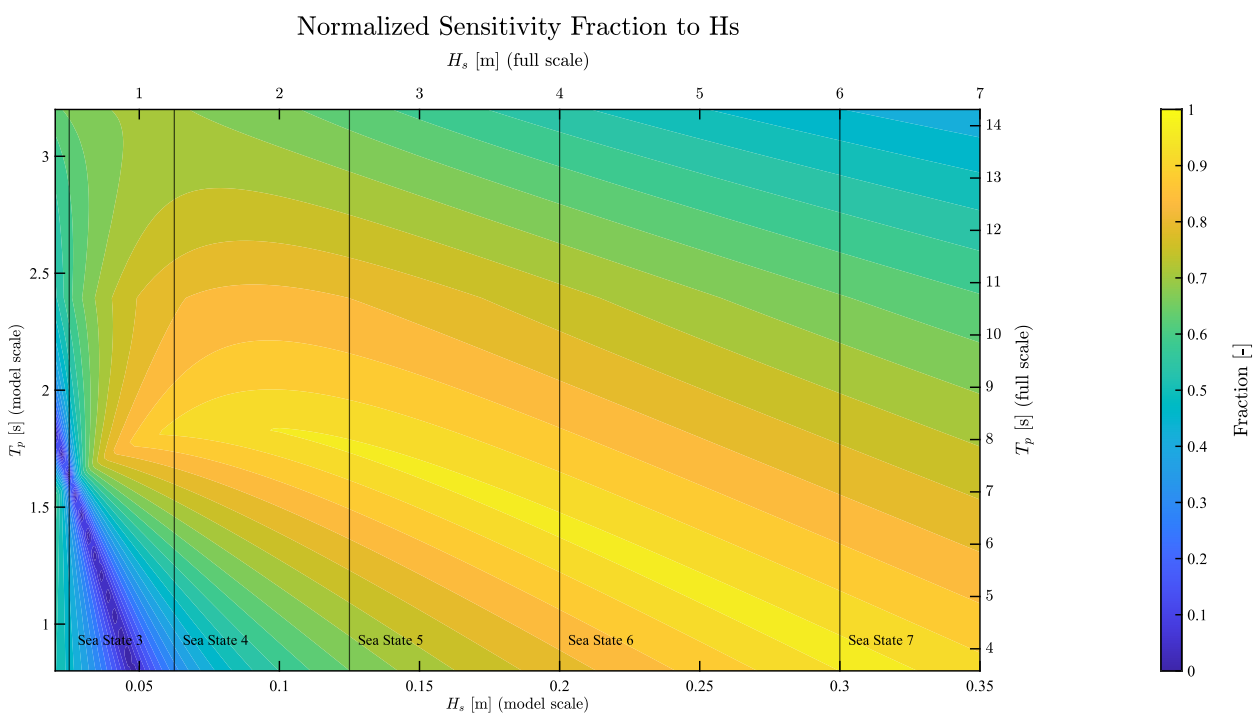

Figure 12: Normalized sensitivity fraction to H_s over the reported (H_s, T_p) domain.

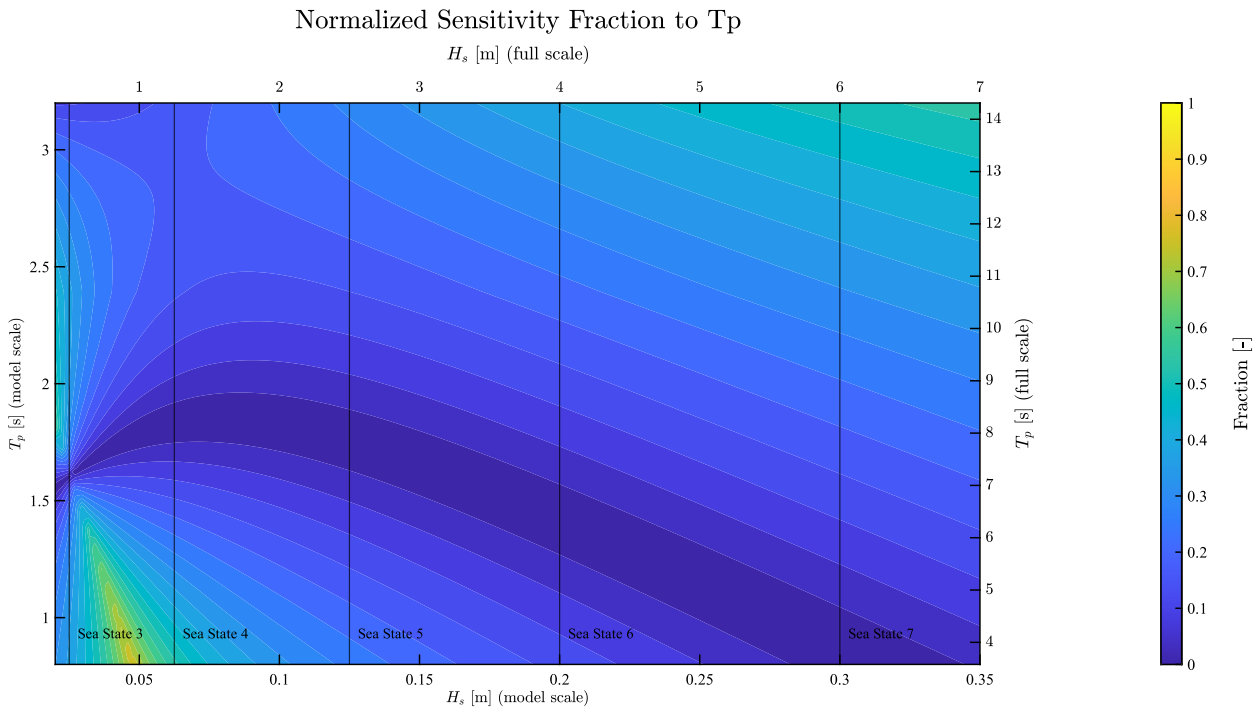

Figure 13: Normalized sensitivity fraction to T_p over the reported (H_s, T_p) domain.

0.35 m and $T_p \approx 0.8$ to 1.0 s in model scale, where the normalized total sensitivity approaches 1. By contrast, the weakest sensitivities occur near the upper-left portion of the domain, roughly $H_s \approx 0.02$ to 0.06 m and $T_p \gtrsim 2.5$ s, where the normalized total sensitivity is close to zero. Thus, the final surrogate is most responsive to input perturbations for larger wave heights and shorter peak periods, while it varies much more weakly for smaller wave heights and longer peak periods.

The variable-level sensitivity maps clarify which inputs drive this pattern. Figure 12 shows that the sensitivity fraction associated with H_s is largest over much of the domain and becomes particularly strong at larger wave heights. In the high-H_s region, the normalized H_s sensitivity approaches 1, indicating that the total sensitivity there is dominated primarily by variation in wave height. In contrast, Figure 13 shows that the contribution from T_p is largest in the complementary region of smaller wave heights and longer periods. In that part of the domain, the normalized T_p sensitivity approaches unity while the H_s contribution weakens, showing that the local structure of the response surface changes from a height-dominated regime to a period-dominated regime as one moves across the (H_s, T_p) plane.

Taken together, Figures 12 and 13 show that the total sensitivity field is not simply large or small; rather, its character changes across the domain. The lower-right region combines large overall sensitivity with strong H_s-dominance, whereas the upper-left region is both

less sensitive in magnitude and more strongly influenced by T_p. This provides a clearer interpretation of the total sensitivity map: it identifies where the surrogate changes most rapidly, while the normalized component maps identify which input is primarily responsible for that variation.

The overlaid sea-state boundaries help place these patterns in physical context. Since the horizontal axis is plotted in model-scale significant wave height, the WMO sea-state bands were converted from full scale to model scale using the same 1:20 Froude scaling used elsewhere in the study. Within the reported domain, the strongest total sensitivity is concentrated toward the larger sea states that remain in range, while the smaller sea states correspond to flatter portions of the response surface. This makes the mathematical sensitivity structure easier to interpret in physically familiar terms.

The same conclusion is reinforced by the term-level sensitivity summary in Table 1. The dominant contributors are the H_s-driven terms. In particular, the linear H_s term and the quadratic H_s^2 term contribute normalized term sensitivities of 0.248 and 0.245, respectively, making them the two strongest retained terms by derivative-based influence. The mixed cubic term $H_s^2 T_p$ is the next most important, with normalized term sensitivity 0.131, followed by the bilinear term $H_s T_p$ at 0.091. By contrast, the purely T_p-driven and direction-coupled terms each contribute substantially less, typically on the order of 0.025 to 0.057. The intercept contributes no derivative-based sensitivity, as expected.

Table 1: Term-level sensitivity summary for the retained PTO total power surrogate. $|\beta|$ is the magnitude of the retained standardized polynomial coefficient. The normalized term sensitivity is the retained term's normalized contribution to the span-scaled total model derivative over the plotted (H_s, T_p) slice.

Term	Term label	$\lvert\beta\rvert$	Mean term total sensitivity	Normalized term sensitivity
1	1	11.170	0.000	0.000
2	T_p	4.974	14.654	0.054
3	H_s	12.668	67.914	0.248
4	$T_p \cos\beta$	2.066	15.698	0.057
5	$T_p \sin\beta$	2.036	6.851	0.025
6	T_p^2	2.061	9.964	0.036
7	$H_s T_p$	3.449	24.829	0.091
8	H_s^2	3.801	66.992	0.245
9	$T_p \sin\beta \cos\beta$	1.107	9.609	0.035
10	T_p^3	0.857	7.324	0.027
11	$H_s T_p^2$	1.367	13.909	0.051
12	$H_s^2 T_p$	1.795	36.008	0.131

An important point from Table 1 is that coefficient magnitude and derivative-based influence

are related but not identical. For example, the intercept has a large coefficient magnitude in absolute value, but its derivative contribution is exactly zero. Conversely, terms such as H_s^2 and $H_s^2 T_p$ exert a strong influence on the sensitivity structure even though their coefficient magnitudes are not the largest in the model. This confirms that the term-level sensitivity analysis provides information that cannot be inferred from coefficient magnitude alone.

Taken together, these results show that the final reduced surrogate has a clear and interpretable sensitivity structure. The response surface is most sensitive in the high-H_s, low-T_p portion of the reported domain, the dominant variable-level contribution shifts between H_s and T_p across the domain, and the dominant term-level contributions arise from the linear, quadratic, and mixed height-period terms. This provides a compact physical and mathematical summary of how the final surrogate responds to changes in the governing inputs.

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